

TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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0.1 CMS Actuation

The tables below cover CMS actuation organized by system: CMDS (Manual, Automatic, Semi-Automatic, Bypass, and Standby modes) and ECM (external pods ALQ-131/ALQ-184 and internal IDIAS).

All CMS actuation described in this chapter is *independent* of the Master Mode (NAV, A-A or A-G) selected.

Table columns are defined as follows:

- **Mode:** The operational context (Master Mode).
- **Direction:** CMS hat direction (Up, Down, Left, Right).
- **Action:** Press type (Short, Long, Long Hold).
- **Function:** What the CMS command activates or controls.
- **Effect / Nuance:** Resulting system behavior, including tactics and constraints.
- **Dash34:** Reference section in the Dash-34 manual.
- **Training:** Recommended BMS training mission(s) (see BMS Training Manual 4.38.1 for descriptions).

0.1.1 CMDS

The ALE-47 CMDS operates in five modes selected via the CMDS CCU MODE knob: Manual (MAN), Automatic (AUTO), Semi-Automatic (SEMI), Bypass (BYP), and Standby (STBY). The program executed in MAN, AUTO, and SEMI modes — Programs 1–4 — is selected via the PRGM knob on the CMDS panel. The PRGM knob has no effect in BYP or STBY.

Only Programs 1–4 are selectable via the PRGM knob. Programs 5 and 6 are independent of the PRGM knob position:

- Program 5 is activated via the cockpit slap switch — outside the scope of CMS — and is not covered in this guide; see Dash-34 § 2.7.2 for details.
- Program 6 is a special-purpose program always available for manual dispensing, independent of the PRGM knob position.

0.1.1.1 Manual Mode

In Manual (MAN) mode, the pilot runs countermeasure programs (1–4 and 6) via **CMS Up** and **CMS Left**.

Table 1: CMS Actuation with CMDS Manual Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS MAN	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once per press. No threat sensing; no automatic triggering. Overrides any AUTO dispensing, if running.	2.7.2.2	18, 28
CMDS MAN	Left	Short	Execute Program 6	Program 6 can be dispensed independently of the PRGM knob position.	2.7.2.2	19

0.1.1.2 Automatic Mode

In Automatic (AUTO) mode, the ALE-47 CMDS dispenses the selected program (1–4) *continuously* in response to threats detected by the Radar Warning Receiver (RWR).

A **CMS Down** press grants consent for AUTO dispensing to start. It will continue until the pilot presses **CMS Right** or expendables are exhausted.

Even in AUTO mode, manual dispensing is still available.

Table 2: CMS Actuation with CMDS Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, interrupting any ongoing AUTO dispensing. After execution, AUTO resumes if the threat persists.	2.7.2.2	18, 28
CMDS AUTO	Left	Short	Execute Program 6	Program 6 is dispensed independently of the PRGM knob position. Overrides AUTO; after execution, AUTO resumes.	2.7.2.2	19
CMDS AUTO	Down	Short	Give Consent; Enable AUTO Dispensing	Grants consent for AUTO dispensing. RWR-detected threats trigger automatic execution of the program (1–4) selected via the CMDS panel PRGM knob. Dispensing continues until the threat clears, the pilot presses CMS Right, or expendables are exhausted. Consent state persists if the pilot switches to MAN mode. If the pilot re-engages AUTO, dispensing resumes immediately on the next detected threat.	2.7.2.1	18, 28
CMDS AUTO	Right	Short	Cancel Consent; Disable AUTO	Removes CMDS consent and places the ALE-47 in Standby. Automatic dispensing halts immediately. The pilot must press CMS Down again to resume AUTO operation.	2.7.2.1	18, 28

0.1.1.3 Semi-Automatic Mode

In Semi-Automatic (SEMI) mode, the CMDS control unit displays DISPENSE RDY and sounds a COUNTER voice message when a threat is detected. A **CMS Down** press grants consent and the ALE-47 CMDS then dispenses *one program*.

If the threat persists or a new threat appears, the CMDS issues COUNTER again.

Even in SEMI mode, manual dispensing is still available.

Table 3: CMS Actuation with CMDS Semi-Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS SEMI	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, independent of RWR threat state. After execution, the CMDS returns to standby and resumes threat detection.	2.7.2.2	18, 28
CMDS SEMI	Left	Short	Execute Program 6	Program 6 is dispensed independently of the PRGM knob position. After execution, the CMDS returns to standby and resumes threat detection.	2.7.2.2	19
CMDS SEMI	Down	Short	Give Consent; Dispense One Program	Dispenses one program (1–4) in response to a COUNTER prompt. If the threat persists or a new threat appears, the CMDS issues COUNTER again. Consent state is retained: if the pilot switches to AUTO, the CMDS begins dispensing immediately on the next detected threat.	2.7.2.1	18, 28
CMDS SEMI	Right	Short	Cancel Consent; Return to Standby	Removes SEMI consent and places the CMDS in Standby. COUNTER messages cease.	2.7.2.1	18, 28

0.1.1.4 Bypass and Standby Modes

BYP and STBY are non-combat modes. Neither supports AUTO or SEMI dispensing; consent state is not tracked in either mode.

Bypass (BYP): In BYP, **CMS Up** dispenses exactly one chaff and one flare per press. The PRGM knob selection has no effect; the dispense quantity is fixed. **CMS Left** dispenses Program 6 normally — Program 6 is available independent of the MODE knob position (see Dash-34 § 2.7.2.2).

Table 4: CMS Actuation with CMDS Bypass Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS BYP	Up	Short	Dispense One Chaff + One Flare	Dispenses exactly one chaff and one flare per press. The PRGM knob selection is ignored. No AUTO or SEMI dispensing occurs in BYP.	2.7.2.1	—
CMDS BYP	Left	Short	Execute Program 6	Program 6 is dispensed independently of the PRGM knob position. Behavior is identical to MAN, AUTO, and SEMI modes.	2.7.2.2	—

Standby (STBY): In STBY, all chaff and flare release is inhibited. No CMS direction produces a dispensing effect. STBY is the only MODE knob position in which CMDS programs and bingo quantities can be edited via the UFC. For program editing procedures, see Dash-34 § 2.7.2.1 and Dash-34 § 2.7.2.3.

0.1.2 CMDS and ECM Joint Consent

When equipped with an external ECM pod, **CMS Down** grants consent to *both* CMDS (in AUTO or SEMI mode) and the external ECM pod *simultaneously*. This joint consent applies to AUTO and SEMI only; BYP does not support AUTO or SEMI dispensing and is not subject to consent authority.

Table 5: CMS Interaction with CMDS and ECM (Consent and Constraints)

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO SEMI + ECM Pod	Down	Short	Joint Consent (CMDS + ECM)	Consent is granted to <i>both</i> the ALE-47 CMDS (AUTO or SEMI mode) and the external ECM pod <i>simultaneously</i> .	2.7.2.1, 2.7.4.2.5	18, 28

0.1.3 Constraints and Edge Cases

State Tracking: In AUTO and SEMI modes, the CMDS retains consent state when the pilot switches to MAN mode. If the pilot re-engages AUTO without pressing **CMS Down**, the CMDS begins dispensing immediately on the next detected threat. BYP and STBY do not have a consent state; switching to either mode does not retain or transfer any prior consent.

Bingo Quantity Behavior: If expendables fall to or below the bingo quantity, the CMDS continues dispensing. A LOW audio message is issued when an expendable reaches the bingo quantity and an OUT audio message is issued when an expendable is depleted. Dispensing does not stop automatically. The pilot must press **CMS Right** to inhibit AUTO dispensing, or switch to MAN mode to take direct control.

RF Switch Override: When ECM consent is active, the ECM system is placed in Standby if the RF switch is moved from NORM to QUIET or SILENT. Returning the RF switch to NORM does *not* restore consent. The pilot must press **CMS Down** again to resume ECM transmission.

IDIAS vs. External ECM Pod: The joint consent described in [Section 0.1.2](#) and in [Table 5](#) applies to external ECM pod configurations *only*. On IDIAS-equipped aircraft, **CMS Left** cycles ECM modes and **CMS Down** has no ECM function.

Ground Safety: On the ground, ECM consent state is Standby. If the pilot holds **CMS Down** on the ground, radiation is emitted until **CMS Down** is released. If ECM must be enabled on the ground, ground personnel must be clear of the radiation area beforehand.